Week 11
Project Studio

Level 1: Introduction

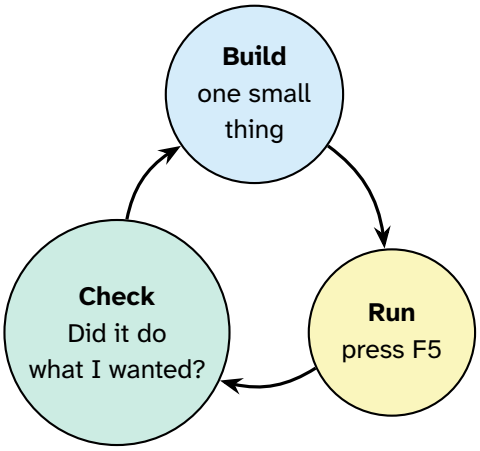
Build, test and improve your project in small steps.

What you will learn

- How real programmers build big programs: one small step at a time.
- How to test a partner's program and give helpful feedback.
- How to finish Milestones M2 and M3 of your art show.

The big idea: build, run, check

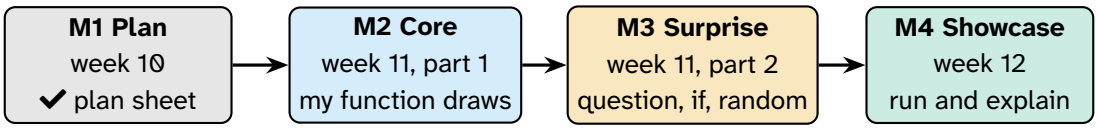
Big programs are not written in one go. Programmers go round this loop all day long:



If something breaks, you know it was the **last small change**.

It is like building a LEGO set bag by bag: finish bag 1, check it looks like the picture, then open bag 2.

Your milestones



Five small steps (the tiny art show from the demo)

1. **Question**: `how_many = int(input("How many bubbles? "))`. Run.
2. **Function**: `def bubble(x, y, color):` and call it once. Run.
3. **Loop**: put the call inside `for i in range(how_many):`. Run.
4. **Random**: random x, y and color. Run twice!
5. **If**: if the number is too big, use a smaller one. Run with `500`.

Tip: Save a version

Before a big change, click **File** → **Save as...** and add `_v2` to the name (`fish_party_v2.py`). If the change goes wrong, version 1 is still safe.

Exercise 11.1: Sprint 1: core drawing works (M2)

Open your project file. Change the example `my_shape` function into **your** function: give it a name that says what it draws, and change the drawing. Draw it once at `(0, 0)`. Run. Then call it a second time somewhere else. Run.

- ☐ My function has a good name: _____
- ☐ It draws my main thing.
- ☐ I call it at least twice, in different places.
- ☐ The program runs with no red errors.

Exercise 11.2: Partner bug hunt

Swap seats with your partner. Run their program **twice**: once with a normal answer, once with a strange one (`0`, a giant number, a word, a capital letter). Write what happened — don't fix it, just report it!

I typed	What happened

Then give **two stars and a wish**:

★ I liked _____

★ I liked _____

I wish _____

Exercise 11.3: Sprint 2: surprise and polish (M3)

Add the rest of your six ingredients. Check off each one in the comment list at the top of your file **when you have seen it work**.

- ☐ My question(s) work (numbers use `int(...)`).
- ☐ My `if` changes something you can see.
- ☐ My loop repeats my function.
- ☐ `random` makes it different every run.
- ☐ A title with `t.write(...)` and my **nickname** as the artist.

Tip: Feedback is a gift

Giving: be kind and specific ("I like how the stars are different sizes"), and make wishes about the program, never the person. **Getting**: say thank you — then **you** decide what to change. It is your art.

Vocabulary box

Iterate To improve something by going round the build–run–check loop again and again.

Milestone A checkpoint on the way to finishing a project.

Playtest Letting someone else try your program to find problems.

Feedback What someone tells you about your work, to help you improve it.

Version A saved copy of your program at one moment, like `_v2`.

Reflection

What bug did you find today? How did you find it?

What feedback did you get? What will you change because of it?

Week 11 Level 1: Introduction

Get Ready for the Showcase

15–20 minutes. This week's homework is project work: finish and practice.

Name (nickname is fine): _____ Date: _____

⭐ Core (everyone): Finish and practice (about 10 minutes)

1. If your project is missing an ingredient, add it now (or write on paper how you will add it next week). Which ingredients still need work?

   _____

2. Practice your **2-minute demo** out loud for someone at home. They are the visitor: let them answer your program's question (or pretend, using your sketch). Use the script in your workbook (chapter 12).
3. Ask your visitor for **two stars and a wish**:

   ⭐ _____

   ⭐ _____

   🪄 _____

⭐⭐ Challenge (want more?): Be ready for strange answers

Visitors sometimes type surprising things. Make your program cope:

- a number that is too big? Use an `if` to make it smaller and print a friendly message;
- `Night` instead of `night`? Add `.lower()` after `input(...)`.

🎨 Creative (make it yours): Make your stand card

Fold a piece of paper in half to make a card that stands up. Write your project's title, your **nickname** as the artist, and one sentence about your artwork. Decorate it! (No real full names, please: nicknames only.)

✅ Self-check: tick each box when it is true

- ☐ My program runs from start to end without red errors.
- ☐ All six ingredients are checked in the comment list at the top of my file.
- ☐ I practiced my 2-minute demo out loud.

☐ I can point to my function, my loop and my `if`.